dnaresilience

Welcome

Example Report

to your DNA Resilience report

Date of birth: 15 Apr 1990	Date reported: 04 May 2021	Sample Number: DNA123
Referring practitioner: Private		

DNA Resilience provides valuable insights into how you can learn to thrive under adverse conditions, using personalised diet, nutraceutical, exercise and lifestyle recommendations to boost your resilience potential and improve your health outcomes



DNALYSIS Biotechnology in collaboration with SUTTONHEALTH

Thoughts on DNA Resilience...



"Choosing DNA Resilience is the first step in your journey towards the realisation of your fullest potential. It means you are consciously deciding on a course of action to successfully adapt to change and overcome the obstacles around you."

So many factors underpin our resilience, including life experiences, the people who guide and support us and our genetic make-up. The influence of genetics in our intellect, emotional responses and behaviours is significant, and driven largely by seven major neurochemicals or hormones. By understanding this complex set of interconnected biological and physiological factors, and supporting the delicate balance between the various parts, we are able to bring to the fore the best version of ourselves. This translates into operating at an enhanced level of adaptability, creativity, motivation, focus, attention, memory, learning, confidence, persistence and optimism. It also means functioning at a significant level of emotional stability, regardless of the circumstances.

This performance panel explores some of the most widely researched genetic factors that influence these primary neurochemical systems and is able to highlight areas of strength and potential hurdles. Regardless of your genetic resilience score, you can excel and in many instances it is those weaknesses that ultimately become an acquired strength. This is a journey where gene–environment interactions and gene–gene relationships shape various realities, and the team of highly skilled practitioners I have put together will help you unlock the giant that lives within each of you.



Richard Sutton

Author: 'The Stress Code: From Surviving to Thriving' and 'StressProof - The Game Plan'

Genetics and personalised medicine

Genes are segments of DNA that contain the instructions your body needs to make each of the many thousands of proteins required for life. Each gene is comprised of thousands of combinations of "letters" (called bases) which make up your genetic code. The code gives the instructions to make the proteins required for proper development and function.

Genetic variations can affect the expression of a gene, thereby affecting metabolic processes that are important for maintaining cellular health and how we respond to environmental interventions such as diet, lifestyle, supplements, and medication.

Knowledge of these genetic variations offers unparalleled insight into your biological systems, allowing your healthcare practitioner to recommend precise interventions aimed at helping you reach your goals and achieve optimal health.



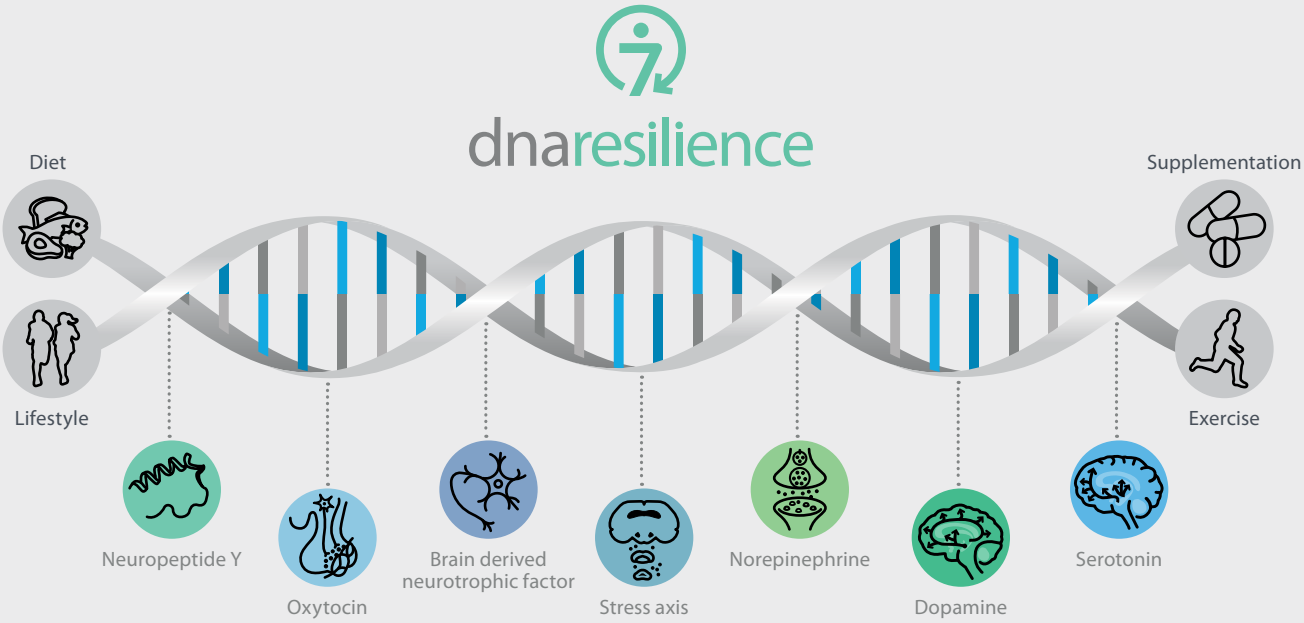
NORMAL GENE
Genotype resulting in baseline potential for disease risk



VARIANT GENE
Genotype resulting in potential for health compromise and need for personalised intervention

Personalised medicine and resilience

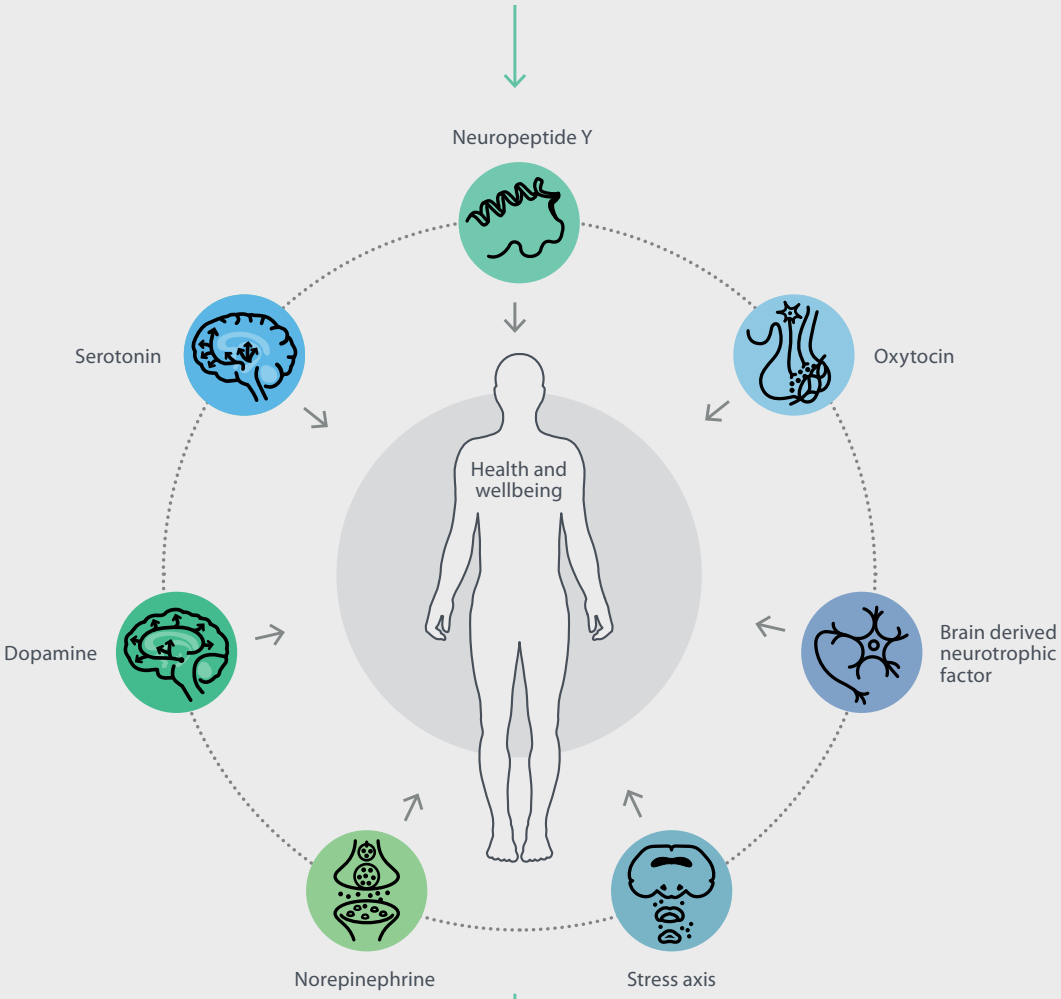
Resilience is the ability to recover quickly from a crisis. Research shows that resilience is highly heritable, meaning there is a strong genetic component to one's baseline resilience and it is how we interact with our environment that determines our ability to thrive under adverse conditions. This report considers the seven molecular areas that underpin our resilience, and provides valuable insight into how you can optimise these areas and boost your resilience using personalised diet, nutraceutical, exercise and lifestyle recommendations.



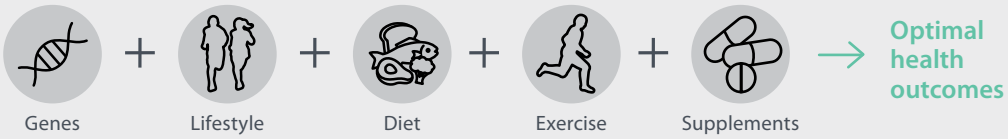
Improving resilience outcomes

DNA Resilience offers insight into the seven molecular areas of resilience in order to help identify strengths and hurdles, as well as recommending personalised interventions which can help you to improve your resilience outcomes and health performance.

DNA RESILIENCE CAN IDENTIFY STRENGTHS AND HURDLES, AND IMPROVE HEALTH PERFORMANCE.



To improve your resilience score and overall health performance:
identify and implement targeted interventions



Result summary

The seven molecular pathways of resilience

The graphs below indicate the extent to which these seven molecular pathways contribute to your own resilience. All of these areas can be strengthened and improved through environmental triggers; pay particular attention to the "Low" and "moderate" results to improve resilience.

MOLECULAR AREA	SCORE
Neuropeptide Y	HIGH
Oxytocin	MODERATE
Brain derived neurotrophic factor	HIGH
Stress axis	LOW
Norepinephrine	HIGH
Dopamine	LOW
Serotonin	LOW

Key areas that require support

If your genotype results are associated with possible weaknesses in key molecular areas that affect resilience outcomes, the molecular areas that require elevated support are indicated. Personalised recommendations for diet, supplementation, exercise and lifestyle, to support these areas, are summarised below.

	DIET	SUPPLEMENT	EXERCISE	LIFESTYLE
Oxytocin	• Focus foods: Magnesium-rich foods (pumpkin seeds), vitamin C-rich foods (kiwi fruit)	• Magnesium • Vitamin C	• Aerobic or circuit style exercise • Yoga	• Aim for 11 or more prosocial acts per week • Ensure adequate debriefing • Healthy physical contact & affection (hugs) • Massage • Loving kindness meditation
Stress axis	• Follow a Mediterranean diet • Avoid refined carbohydrates and processed sugars • Limit caffeine & alcohol • Focus foods: Magnesium-rich foods (pumpkin seeds), omega 3 & zinc-rich foods (fatty fish & shellfish), green tea & cacao	• Vitamin D • Zinc • Omega 3 fatty acids • Magnesium • N-acetylcysteine • Adaptogens	• A programme that includes: - Regular exercise: swimming & yoga - High intensity training	• Weight management • Vagal activation: - Meditation - Deep breathing exercises - Cold water immersion • Spend at least 15 minutes per day in natural environments • Relaxing music (preferably light classical)
Dopamine	• Follow a Mediterranean style diet • Avoid refined carbohydrates, processed sugars & fried foods • Practice mindful eating • Eliminate caffeinated foods & beverages • Limit or avoid alcohol • Focus foods: Cruciferous vegetables (broccoli), magnesium-rich foods (pumpkin seeds), flax, vitamin B-rich foods (green leafy vegetables, eggs)	• N-acetylcysteine • SAMe • Vitamin D3 • Vitamin B complex with methylfolate • Magnesium • Sulforaphane	• Aerobic exercise for 30 minutes at least three times a week	• Limit exposure to environmental toxins • Meditation & yoga • Maintain strong social connections & relationships • Practice 'reframing' during stressful periods
Serotonin	• Follow a Mediterranean diet • Consider intermittent fasting or caloric restriction several times a week (between 2-3) • Focus foods: Tryptophan-rich foods (chicken, turkey), magnesium-rich foods (pumpkin seeds), manganese-rich foods (pineapple), omega 3-rich food (fatty fish)	• Omega 3 fatty acids • Magnesium (glycinate or L-threonate) • Vitamin D • Vitamin B complex including methylfolate • Adaptogens	• A programme that includes: - Moderate intensity aerobic exercise for 20-40 minutes at least 3 times per week - High intensity training - Yoga	• Meditation and yoga • Maintain strong social connections • Practice 'reframing' in stressful periods • Frequent nature exposure • Massage & physical therapies • Learning and acquiring new skills (sports, language, complex tasks) • Listen to relaxing & enjoyable music

Always consult your healthcare practitioner before embarking on a new eating plan or intense exercise programme.

Your results and recommendations

Neuropeptide Y

Your results

Your NPY contribution to resilience

LOW — MODERATE — HIGH

Normal molecule

Altered molecule function due to a gene variant

GENE NAME	GENE VARIATION	RESULT	GENE IMPACT
NPY	-399 C>T	TT	

NPY molecule influence and your outcomes

NPY protects against stress induced depression, worry and anxiety, and helps one to perform better mentally and physically under pressure. It is involved in pain reduction by counteracting the primary stress hormones (cortisol and adrenaline), as well as lowering blood pressure, heart rate and inflammation under stressful and pressurised conditions. NPY decreases brain sensitivity to leptin, which then exerts an influence on obesity by creating new fat cells - thus promotes fat storage.

STRENGTHS

Those with the TT genotype have good protection against the negative effects of stress (i.e. anxiety and depression); perform well mentally and physically under pressure, and show good immune and hormonal regulation under stressful conditions.

HURDLES

You may struggle with weight control under periods of stress and high demands.

Oxytocin

Your results

Your Oxytocin contribution to resilience

LOW — **MODERATE** — HIGH

Normal molecule

Altered molecule function due to a gene variant

GENE NAME	GENE VARIATION	RESULT	GENE IMPACT
OXTR	G>A	AG	↑

OXTR molecule influence and your outcomes

The oxytocin receptor is central in inhibiting the brain's primary fear and stress centre - the amygdala, and promotes courage, calmness, empathy, optimism, trust, self-belief and self esteem. It promotes anti-inflammatory and antioxidant activity and also lowers blood pressure and cortisol.

OXTR stimulates growth hormone and insulin-like-growth-factor (IGF-1), which helps with cell repair and growth. It also helps increase serotonin and BDNF, which promotes mood stability and adaptability.

STRENGTHS

Isolation and solitude do not seem to take the same toll on you as the GG genotype. This may be an advantage in this period of rolling lockdowns and social distancing.

HURDLES

Stress and protracted pressure can increase your risk of mood disorders such as depression and anxiety. There may be more difficulty in overcoming historic life traumas, and there is a higher risk of developing stress-related health issues.

OXTR recommendations

DIET

- Normal stress and performance principals apply. Eat a good variety, lots of fresh vegetables and fruits, seeds, legumes, fish, poultry, olive oil and unrefined hypoallergenic grains (rice, millet, quinoa etc.)
- Focus foods include:
  - Green vegetables, nuts and seeds, oat bran
  - Citrus, kiwi fruit, guavas
  *(**Note:** if you have intolerance to any of these foods e.g. diverticulitis, rather opt for supplementation)*

LIFESTYLE

- Make a conscious effort to show more compassion, care, support and empathy. If possible, aim for 11 or more prosocial acts every week
- Share your stress experiences with those close to you. In simple terms - open up!
- Try to engage in more physical contact with those who are close to you. There are tiny pressure receptors located throughout the body that convert the affection into oxytocin. If you are not a touchy-feely person, it just means that you should lean towards the other strategies more strongly
- Massage or physical therapies such as osteopathy
- Loving kindness meditation. This form of meditation is exceptionally powerful in promoting resilience through oxytocin.

EXERCISE

- 10 minutes, or longer of aerobic or circuit style exercise
- Yoga (30 minutes or more). Note that if you are pressed for time a high quality 15 minute session will help

SUPPLEMENTS

- There are a few supplements that can up regulate oxytocin. These include:
  - Magnesium
  - Vitamin C

Brain derived neurotrophic factor (BNDF)

Your results

Your BDNF contribution to resilience

LOW — MODERATE — **HIGH**

Normal molecule

Altered molecule function due to a gene variant

GENE NAME	GENE VARIATION	RESULT	GENE IMPACT
BDNF	Val66Met	CC	

BNDF molecule influence and your outcomes

BDNF is involved in enhancing memory, augmenting reading and the retention of information, and facilitates creativity and innovation. It has a strong influence in cognitive ability and potential and offers protection against anxiety and depression.

STRENGTHS

Those with the CC genotype show good potential to cope with challenges and adversity, have strong protection against stress-related mental health compromise and exhibit a sound cognitive base.

HURDLES

Chronic stress can reduce BDNF production by up to 40%. Weight gain compromises BDNF production, and advancing age reduces production.



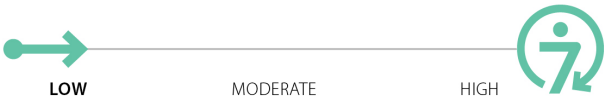
Stress axis

The stress axis plays a central role in the regulation of adrenal hormones. An increase in glucocorticoid levels in response to stressors is advantageous in enabling the body to restore homeostasis. However, if glucocorticoid levels remain chronically elevated due to exposure to chronic stress, this in turn increases risk for developing an impaired stress axis, which is a strong risk factor for inflammatory diseases. The stress axis score is comprised of the analysis of variants on two genes: FKBP5 and CRHR.



Your results

Your stress axis contribution to resilience



Normal molecule

Altered molecule function due to a gene variant

GENE NAME	GENE VARIATION	RESULT	GENE IMPACT
FKBP5	C>T	TT	↑
	A>C	CC	↑
CRHR	C>T	CT	↑
	G>T	GT	↑



Your FKBP5 molecule influence and outcomes

FKBP5 has a structural and functional effect on the brain's fear centre, and influences the overall stress responses (regulates or dysregulates). It sustains the stress axis (blocks down-regulation, influences the production of cortisol in response to stressful events, and influences emotional responses to stress. It has an influence in pain responses, and influences blood sugar levels and weight.

STRENGTHS



Those who carry the FKBP5 T and C variants can positively influence and alter the molecule in the long term through proactive measures.

HURDLES



Stressful events or conditions can be associated with worry and anxiety, low mood or depression, rigidity, and reduced adaptability, and raised systemic inflammation and immune disorders.

FKBP5 recommendations

DIET

- Avoid processed sugars
- Consider a lectin avoidance diet
- Limit alcohol - but most importantly, avoid binge drinking
- Reduce intake of coffee
- Take care to assess food intolerance so as not to induce additional stress within the system
- Include more vegetables and seasonal fruit as well as wild fish and shellfish, seeds and hypoallergenic grains (rice, quinoa, millet, oats, sorghum, tapioca)

LIFESTYLE

- Try to get your weight within healthy ranges as FKBP5 is principally expressed in fat cells. The lower one's body fat, the less of this protein one is likely to express
- Vagal activation. Activities that stimulate and strengthen this system are invaluable due to the fact that the vagus nerve is able to 'turn off' the stress axis. Daily exposure to one or more of these activities is strongly encouraged: meditation; breathing exercises; cold immersion (water); swimming; osteopathy (especially visceral); acupuncture, reflexology and other physical therapies
- Spending 15 or more minutes in natural environments
- Relaxing music (preferably light classical)

EXERCISE

- Exercise (aerobic or resistance) is always valuable during stressful periods in ones life. This said, the most effective forms of exercise are those that are able to promote greater biological stability through strong activation of the vagus nerve (to find out more get a copy of The Stress Code). These include:
  - Swimming
  - Yoga

SUPPLEMENTS

- There are many nutritional supplements that have a positive effect on FKBP5 expression and levels. These include:
  - N-acetylcysteine (very effective)
  - Vitamin D (effective)
  - Zinc
  - Magnesium (effective)
  - Omega 3 fatty acids (very effective)

CRHR molecule influence and your outcomes

CRHR plays a critical role in the stress response (HPA axis) and its intensity; it amplify the experience of stress, including the perception of fear. It influences emotional state and responses to stress, as well as cognition, especially memory and learning, and plays a role in immune behaviours within the digestive tract.

STRENGTHS

For individuals with the CRHR C and G variants, this molecule can be positively influenced and altered in the long term through proactive measures. There is also a degree of protection against stress-induced digestive disorders (consider that there are numerous factors that could overshadow this benefit).

HURDLES

Greater fear responses to demanding and challenging circumstances, disproportionately high-stress responses, the predisposition towards experiencing anxiety and/or depression in response to stress, and compromised memory and learning under stressful conditions.

CRHR recommendations

DIET

- Avoid processed sugars
- Limit alcohol - but most importantly avoid binge drinking
- Reduce intake of coffee and other stimulants
- Take care to assess food intolerances so as not to induce additional stress within the system
- Include more vegetables and seasonal fruit as well as wild fish, shellfish, seeds and hypoallergenic grains (rice, quinoa, millet, oats, sorghum, tapioca)
- Include green tea and cacao or dark chocolate from time to time

LIFESTYLE

- Reframing: Viewing stress and challenges in a positive light and as a growth opportunity can have immense biological advantages. To find out more get a copy of The Stress Code
- Vagal activities are the single most important regulator measure. These include: meditation; breathing exercises; cold immersion (water); swimming; osteopathy (specially visceral); acupuncture, reflexology & other physical therapies
- Spending 15 or more minutes in natural environments
- Relaxing music (preferably light classical)
- Focus on measures and techniques that target oxytocin. This can be achieved through several approaches:
  - Showing more compassion, care for others, support and empathy. If possible aim for 11 or more prosocial (moral) acts every week.
  - Physical contact with those who are close to you Massage or physical therapies such as osteopathy, chiropractics and physiotherapy
  - Loving kindness meditation

EXERCISE

- High intensity exercise (resistance or aerobic) that raises lactic acid has been shown to inhibit the release of stress hormones
- Swimming (due to vagus nerve influence)
- Yoga (due to vagus nerve influence)

SUPPLEMENTS

- There are many nutritional supplements that have a positive effect on CRHR1 expression and levels. These include: zinc, magnesium, vitamin C, omega 3 fatty acids, adaptogens *(under the supervision of your healthcare provider)*

Norepinephrine

Your results

Your norepinephrine contribution to resilience

LOW — MODERATE — **HIGH**

Normal molecule

Altered molecule function due to a gene variant

GENE NAME	GENE VARIATION	RESULT	GENE IMPACT
DBH	C>T	CC	

Norepinephrine molecule influence and your outcomes

DBH encodes the enzyme, dopamine beta (β)-hydroxylase, which converts dopamine to norepinephrine. It promotes focus and sustained attention, reduces the risk of stress-associated depression and anxiety, reduces aggression, and impulsiveness, and protects against neurosis.

STRENGTHS

Individuals with the CC genotype show strong emotional footing during challenging and demanding circumstances, good cognitive aptitude during stressful periods, and have a lower risk of stress-induced headaches and migraines as well as other physical symptoms.

HURDLES

Yeast, fungal or bacterial infections may offset this resilience advantage. If you are struggling with anxiety, depression, aggressive outbursts, neurosis, headaches, and poor attention, it may be worthwhile considering an Organix Acid Test available through Nordic Laboratories.



Dopamine

Dopamine is an excitatory neurotransmitter that is synthesized in the brain, and is responsible for modulating reward and pleasure. It is also essential for optimal cognitive function, as well as working memory. Dopamine is an important contributor to the generation of the stress response and coping responses, where environmental stressors, both physical and psychological, activate the brain dopaminergic circuits, which lead to acute and chronic changes in neuronal functions. The dopamine score is comprised of the analysis of variants on three genes: DRD2, DRD4 and COMT.



Your results

Your dopamine contribution to resilience



Normal molecule

Altered molecule function due to a gene variant

GENE NAME	GENE VARIATION	RESULT	GENE IMPACT
DRD2	Taq1A	TC	↑
DRD4	-521 T>C	TT	↑
COMT	Val158Met	AA	↑



DRD2 molecule influence and your outcomes

DRD2 encodes a receptor for dopamine. It plays a role in positively influencing sleep, promoting memory, and optimising learning, attention, focus, and intellectual agility, as well as emotional stability. It is central in motivation, and drives performance, especially in stressful situations. It also influences pain perception.

STRENGTHS



Those with the T variant are responsive to cognitive training and are motivated by positive rewards.

HURDLES



Stress may trigger the overconsumption of food. There is a greater inclination for cravings for sugars, carbohydrates, and fats. Significant traumas may be associated with PTSD. One may be impulsive and thrill-seeking and prone to novelty-seeking behaviour.

DRD2 recommendations

DIET

- Avoid all refined carbohydrates, processed sugars, and fried foods - especially during stressful periods
- Focus on maintaining stable blood glucose levels
- Practice mindful eating
- During stressful events it may be wise to manage portion control at meals
- Ensure good overall food variety. Lots of fresh vegetables (preferably organic) and seasonal fruits, seeds, legumes, wild fish, free range poultry, cold pressed olive oil and unrefined, hypoallergenic starches (rice, millet, quinoa, sweet potato etc.)

LIFESTYLE

- Try to engage in regular meditation, specifically focused based. Several times a week for 15-30 minutes is suggested
- Listen to enjoyable music daily (30-45 minutes)
- Maintain strong social connections and relationships
- Practice 'reframing' during stressful periods (see Stress Code for details)
- Prioritise health during stressful periods

EXERCISE

- Aerobic exercise for 30 minutes at least three times a week is strongly recommended.
  - Running, cycling, walking, swimming, hiking, steps, circuit weight training

SUPPLEMENTS

- There are many nutritional supplements that have a positive effect on DRD2 expression and levels. These include:
  - Omega 3 fatty acids
  - Vitamin D3
  - SAMe
  - Vitamin B complex containing methylfolate
  - Probiotics (specifically L. Plantarum)
  - Adaptogens *(should not be taken without supervision by your healthcare provider)*

DRD4 molecule influence and your outcomes

DRD4 encodes a dopamine receptor, which has a major influence on dopamine levels within the brain. It regulates emotions and complex behaviours and has a strong resilience influence. It promotes cognition, augments memory and learning, enhances movement potential, and modulates overall neurochemical and hormonal balance.

STRENGTHS

Individuals with the TT genotype are responsive to environmental input. This molecule can be positively influenced and altered in the long-term through proactive measures.

HURDLES

Under stressful and demanding conditions the following may occur; depression, impaired memory, novelty seeking behaviour, challenges with measuring emotional responses and short temper and difficulty in adapting under stressful circumstances. One can also be impulsive and thrill-seeking.

DRD4 recommendations

DIET

- Increase intake of tyrosine rich foods. These include: wild fish, free range chicken, wild rice, sweet potato, avocado, seeds, cacao/cocoa
- Caffeinated beverages:
  - Green tea
  - Coffee (no more than 3 cups) **take note of COMT*
- Other focus foods: dark green leafy vegetables, fatty fish such as salmon and sardines, lentils, blueberries

LIFESTYLE

- Meditate! Use focused based meditation daily
- Enjoy a massage or physical therapy. Massage can increase dopamine by around 30% in a very short period of time
- Listen to enjoyable music daily (30-45 minutes)

EXERCISE

- Aerobic exercise for 30 minutes at least three times a week is strongly recommended
  - Running, cycling, walking, swimming, hiking, steps, circuit weight training

SUPPLEMENTS

- There are many nutritional supplements that have a positive effect on DRD4 expression and levels. These include:
  - Omega 3 fatty acids (can increase dopamine by as much as 40%)
  - Methylfolate
  - Vitamin D3
  - Probiotics (specifically L. Plantarum)
  - Resveratrol
  - Curcumin
  - Adaptogens *(under the supervision of your healthcare provider)*

COMT molecule influence and your outcomes

COMT encodes an enzyme responsible for the break-down of dopamine, apinephrine and norepinephrine. It is involved in promoting mood stability, adaptability, cognitive ability, and motivation and augments movement. It also plays a role in blood vessel, immune, and hormonal regulation.

STRENGTHS

With a 75% reduction in enzyme activity in AA genotype carriers, dopamine and norepinephrine are likely to be elevated, even under non-stress conditions. This can lend itself to exceptional cognitive and executive capabilities (especially under non-stress conditions), and a results-driven mindset.

HURDLES

Due to excessive dopamine, norepinephrine, and epinephrine (especially in the executive regions) the following issues may occur under stressful conditions; Lower cognitive ability, negative mood states that can include low mood, worry, and despair, and exaggerated stress responses as well as sustained heightened stress responses.

COMT recommendations

DIET

- Increase intake of cruciferous vegetables such as: broccoli, cauliflower and cabbage
- Include flaxseed in your diet
- Eliminate caffeinated foods and beverages such as: cola (diet and normal), coffee
- Limit or avoid alcohol
- Avoid:
  - Processed foods, fried foods, refined sugars
  - Pesticides in food (choose organic when possible)
  - Unfiltered water
  - Heavy metals such as mercury (found in tuna and other large species of fish)

LIFESTYLE

- Meditation and deep breathing
- Manage weight and keep body fat percentage in a healthy range
- Focus on good sleep hygiene to get enough sleep (8 hours per night)
- Regular saunas and steam bath
- Support liver detoxification through the avoidance of:
  - Parabens (found in cosmetics), xenoestrogens (found in plastics, pesticides, chemicals)
  - Pollution (where possible)
  - Household chemicals
  - Smoke

EXERCISE

- Aerobic exercise for 30 minutes at least three times a week is strongly recommended.
  Options include:
  - Running, cycling, walking, swimming, hiking, steps, circuit weight training, tennis, soccer etc.

SUPPLEMENTS

- There are many nutritional supplements that support COMT activity and have a positive effect on dopamine levels. These include:
  - N-acetylcysteine
  - SAMe
  - Vitamin D3
  - Methylfolate (vitamin B9), vitamin B2, B6 and B12
  - Magnesium
  - Sulforaphane



Serotonin

Serotonin is primarily found in the gastrointestinal tract, as well as blood platelets, and the central nervous system (CNS) and is an important modulator of mood, contributing toward feelings of well-being and happiness. Low levels of serotonin are associated with mood disorders, including anxiety and depression. The serotonin score is comprised of the analysis of variants on four genes: TPH2, HTTLPR, HTR1A, and HTR2A.



Your results

Your serotonin contribution to resilience



Normal molecule

Altered molecule function due to a gene variant

GENE NAME	GENE VARIATION	RESULT	GENE IMPACT
TPH2	-844 G>T	TT	○
HTTLPR	A>C	AC	↑
HTR1A	-1019 C>G	CG	↑
HTR2A	-1438 G>A	GA	↑



TPH2 molecule influence and your outcomes

TPH2 encodes an enzyme that is responsible for making serotonin in the brain. It promotes mood and emotional stability, enhanced planning and decision making, enhanced sensory gating (preventing irrelevant information from overwhelming the brain), and regulation of behaviour and impulsivity as well as cognitive function including concentration.

STRENGTHS



Those with the T variant have protection against stress-related anxiety (due to raised GABA and BDNF), and a lower risk of stress-related depression (BDNF).

HURDLES



There may possibly be reduced serotonin levels within the brain, however, this is dependant on multiple factors, and should be considered in context. Lower serotonin is linked to difficulty in developing new skills, challenges with long-term planning, indecisiveness, and higher reactivity to emotional environments, as well as less social inclination and a reduced ability to filter sensory information.

HTTLPR molecule influence and your outcomes

HTTLPR encodes for the serotonin transporter (SERT), which is a protein that functions to transport serotonin. It promotes successful adaptation to stressful and demanding circumstances, memory and cognition, positively influences emotional stability and integrity, and is associated with regulating social behaviours. It also enables good connectivity and functionality within a region of the brain known as the visual-limbic subnetwork.

STRENGTHS

The aging process appears to offset some of the limitations of underexpression associated with the A variant.

HURDLES

Individuals with the A variant may present with reduced resilience, a higher risk of depression and anxiety in response to stress, be more prone to emotional suppression and dysregulation, less inclined to seek support or assistance during stressful events, have possible limited responsiveness to some serotonin reuptake inhibitors (SSRI's), and exhibit lower overall BDNF expression.

HTTLPR recommendations

DIET

- The Mediterranean diet offers good support in this regard. This would encompass good overall variety, lots of fresh vegetables (preferably organic) and seasonal fruits, seeds, legumes, wild fish, poultry cold pressed olive oil and unrefined hypoallergenic starches (oats, rice, millet, quinoa, sweet potato etc.)
- Intermittent fasting several times a week provided you don't have any underlying health conditions or train intensely **Remember to always consult your healthcare practitioner before embarking on a new eating plan*

LIFESTYLE

- Nature exposure (especially sunlight) for a collective of 2-hours a week
- Learning and acquiring new skills (sports, language, complex tasks)
- Maintain strong social connections and relationships
- Try to avoid pollution (environmental and food)
- Practice loving kindness meditation as frequently as possible

EXERCISE

- Moderate intensity aerobic exercise (running, swimming, cycling, walking, tennis etc.) for a period exceeding 20-40 minutes. Ideal frequency is 3-5 times a week
- High intensity training. Short intense sessions can be a good way to boost BDNF and serotonin. Make sure that you get your doctors sign-off and don't perform more than 2-3 times a week due to fatigue and potential injuries
- Yoga

SUPPLEMENTS

- There are many nutritional supplements that have a positive effect on SERT expression, Oxytocin or BDNF. These include
  - Adaptogens *(should not be taken without supervision by your healthcare provider)*:
  - Omega 3 fatty acids
  - Magnesium
  - Vitamin C

HTR1A molecule influence and your outcomes

This receptor promotes successful adaptation to stressful and demanding circumstances, and increased adaptability within the brain under stressful conditions (neuro-plasticity). It protects against and moderates aggressive behaviours, impulsivity, stress related depression, anxiety and panic disorders. It also positively influences several major hormones and biological systems including the vagus nerve (see stress axis), oxytocin, growth hormone, endorphins and adrenocorticotropic hormone (ACTH), which plays a central role in the release of cortisol.

STRENGTHS

In individuals with the HTR1A G variant, this molecule can be positively influenced and altered in the long term through proactive measures.

HURDLES

Stressful events or conditions can be associated with the following; reduced resilience potential, risk of stress-related emotional disorders including depression, anxiety, aggression, compulsive behaviours, panic disorder, and eating disorders, a reduced ability to effectively regulate stress responses, neurochemical and hormonal disruptions, especially during stressful events, and reduced connectivity within key circuits of the brain.

HTR1A recommendations

DIET

- The Mediterranean diet offers great support. This encompasses good overall food variety, lots of fresh vegetables (preferably organic) and seasonal fruits, seeds, legumes, poultry, wild fish, cold pressed olive oil and unrefined hypoallergenic starches (rice, millet, quinoa, sweet potato etc.)
- Focus foods:
  - Tryptophan-rich foods: chicken, turkey, beef (in moderation), organic tofu, fish (salmon), edamame beans, squash and pumpkin seeds, oatmeal, eggs
  - Manganese-rich foods: Organic tofu, sweet potato, pine nuts, brown rice, chickpeas, lima beans, pineapple

LIFESTYLE

- Maintain strong social connections and relationships
- Treat yourself to physical therapies. These include massage, acupuncture, osteopathy, reflexology etc.
- Engage in activities that make you happy and bring fulfilment
- Listen to relaxing and enjoyable music
- Practice meditation or yoga frequently
- Get outdoors for at least 2 hours a week

EXERCISE

- Regular aerobic exercise (cycling, running, swimming, walking, hiking, tennis, soccer etc.) is important 3-5 moderate intensity sessions a week is ideal, for a period of 20-40 minutes. This would depend on fitness levels and general health

SUPPLEMENTS

- There are many nutritional supplements that have a positive effect on HTR1A and/or serotonin expression. These include: omega 3 fatty acids; magnesium; vitamin D; vitamin B complex including methylfolate; adaptogens *(should not be taken without supervision by your healthcare provider)*

HTR2A molecule influence and your outcomes

This serotonin receptor assists in successful adaptation of the brain to stressful and demanding circumstances, the formation of new brain cells (neurogenesis), enhances cognition, creativity and adaptability and learning, is instrumental in long term emotional integrity, and promotes openness. In an over expressed and activated state, it may cause anxiety, pessimism and depression.

STRENGTHS

Individuals who have the A variant may show exceptional potential for learning, good cognitive flexibility and overall cognitive/intellectual aptitude, a creative disposition, and the ongoing capacity to adapt (impacting overall resilience potential).

HURDLES

There is an increased risk of stress-related anxiety and depression, a higher propensity for stress-related neurosis and pessimism, and poor sleep patterns under stressful conditions. Carriers of the A variant may be prone to chronic fatigue and less sociable overall. They also exhibit reduced BDNF expression.
Note that many of the advantages in executive functionality may be overshadowed/offset by emotional instability and fatigue.

HTR2A recommendations

DIET

- The Mediterranean diet offers great support. This encompasses good overall food variety, lots of fresh vegetables (preferably organic) and seasonal fruits, seeds, legumes, poultry, wild fish, cold pressed olive oil and unrefined hypoallergenic starches (rice, millet, quinoa, sweet potato etc.)
- Focus foods:
  - Tryptophan-rich foods: chicken, turkey, beef (in moderation), organic tofu, fish (salmon), edamame beans, squash and pumpkin seeds, oatmeal, eggs
  - Manganese-rich foods: organic tofu, sweet potato, pine nuts, brown rice, chickpeas, lima beans, pineapple

LIFESTYLE

- Maintain strong social connections and relationships
- Treat yourself to physical therapies. These include massage, acupuncture, osteopathy, reflexology etc.
- Engage in activities that make you happy and bring fulfilment
- Listen to relaxing and enjoyable music
- Practice meditation or yoga frequently
- Get outdoors for at least 2 hours a week

EXERCISE

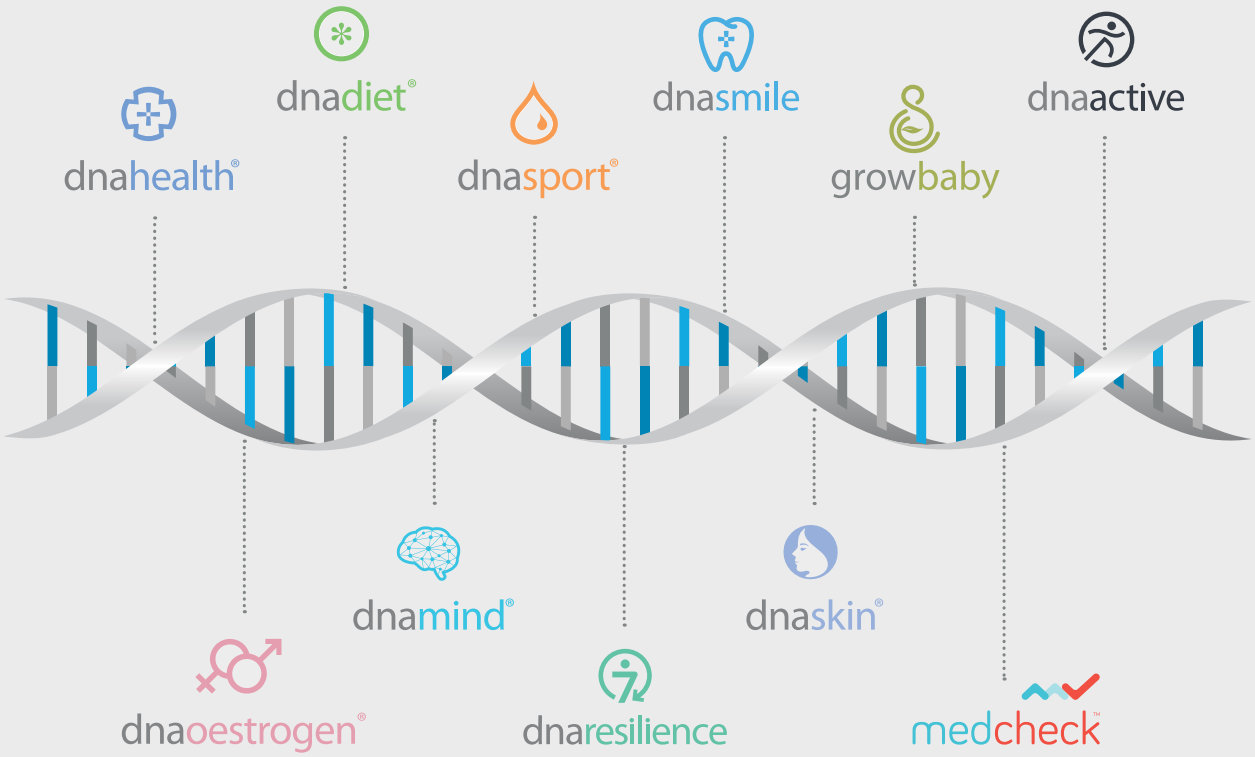
- Regular aerobic exercise (cycling, running, swimming, walking, hiking, tennis, soccer etc.) is important. 3-5 moderate intensity sessions a week is ideal, for a period of 20-40 minutes. This would depend on fitness levels and general health

SUPPLEMENTS

- Omega 3 fatty acids (improves HT1A receptor function, which can balance HT2A)
- Magnesium (improves HT1A receptor function, which can balance HT2A)
- Chromium (modulates HT2A)
- Adaptogens *(should not be taken without supervision by your healthcare provider)*

A lifetime of optimal health awaits you

Your genes do not change, which means our laboratories will only ever need one cheek swab sample from you. Throughout your life, as your health goals and priorities change, we can continue to provide valuable health insights from this single cheek swab to support your unique health journey.



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Risks and Limitations:
DNAlysis Biotechnology has a laboratory with standard and effective procedures in place for handling samples and effective protocols in place to protect against technical and operational problems. However as with all laboratories, laboratory error can occur; examples include, but are not limited to, sample or DNA mislabelling or contamination, failure to obtain an interpretable report, or other operational laboratory errors. Occasionally due to circumstances beyond DNAlysis Biotechnology's control it may not be possible to obtain SNP specific results.